Slides week 21

Algebraic Structures

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Deep Dive Slides

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Deep Dive Slides

- These slides are similar to the lecture slides but include added motivations and explanations to enhance your learning experience.
- They are self-contained for independent study and do not provide additional material.
- Optional proofs are included (marked as optional).
- If you prefer a straightforward approach, you can study from the lecture slides without missing anything.

Week 21: Goals

Last time:

- New algebraic structures: Fields!
- Subfields
- Quick Subfield Theorem
- Homomorphism of rings, ID and fields

Today:

- Isomorphisms of rings, ID and fields
- Ideals

Reminder: Ring Homomorphism, ID Homomorphisms and field homomorphisms

Next

- We will start recalling the definition of **ring homomorphisms**.
- Then, we will recall the definition of **ID homomorphisms** and **field homomorphisms**.
- We know that integral domains and fields are in particular rings, but they have more axioms
- Thus, an **ID homomorphism** is the same as ring homomorphisms, but between integral domains.
- Similarly, a **field homomorphism** is a ring homomorphism between fields.

Ring homomorphism

Definition

Let R and S be rings. A **ring homomorphism** from R to S is a mapping

$$\theta : R \to S$$

that satisfies:

- $\theta(a+b) = \theta(a) + \theta(b)$, and
- $\theta(ab) = \theta(a)\theta(b)$,

for all $a, b \in R$.

Remark

Recall that

$$\text{Fields} \subset \text{Integral Domains} \subset \text{Rings}.$$

Thus

- an **integral domain homomorphism** is just a **ring homomorphisms** between two integral domains.
- Similarly, a **field homomorphism** is just a **ring homomorphisms** between two fields.

An example

Complex conjugation

- In the next slides, we are going to show that complex conjugation is a ring homomorphism.
- Recall the the conjugate of a complex number $a + ib$ is $a - ib$.

Example

Complex conjugation map $f : \mathbb{C} \to \mathbb{C}$ given by

$$f(a+ib) = a - ib.$$

Let us show that this is a ring homomorphism.

- Conjugation preserves sum:

$$\begin{aligned} f\left((a+ib)+(c+id)\right) &= f\left((a+c)+i(b+d)\right) \\ &= (a+c)-i(b+d) \\ &= (a-ib)+(c-id) \\ &= f(a+ib)+f(c+id). \end{aligned}$$

Example: complex conjugation - Part 2

Example

- Conjugation preserves multiplication:

$$\begin{aligned} f\left((a+ib)\cdot(c+id)\right) &= f\left((ac-bd)+i(ad+bc)\right) \\ &= (ac-bd)-i(ad+bc) \\ &= (a-ib)\cdot(c-id) \\ &= f(a+ib)\cdot f(c+id). \end{aligned}$$

Remark

- Since $\mathbb{C}$ is an integral domain, the map f is also an **integral domain homomorphism**.
- Since $\mathbb{C}$ is a field, the map f is also a **field homomorphism**.

Ring isomorphisms

- In Group Theory, a group isomorphism is a map that is at the same time bijective and a group homomorphism.
- Similarly, for rings, we say that a map is **ring isomorphism** if it is at the same time bijective and a ring homomorphism.
- As for groups, if there is a ring isomorphism $\varphi : R \to S$, then R and S are essentially the same ring.
- In particular, if there is a ring isomorphism $\varphi : R \to S$, then R and S share the same algebraic properties. E.g
  - R have the same cardinality,
  - if R has unity, so has S,
  - if R is commutative, so is S, etc.

Definition.

Let R and S be rings. An **isomorphism** from R to S is a mapping

$$\theta : R \to S$$

$\forall s \in S$, $\exists r \in R$ such that $\theta(r) = s$

that is **injective**, **surjective**, and a **ring homomorphism**.

$\theta(x) = \theta(y) \Longrightarrow x = y$

$\theta(x + y) = \theta(x) + \theta(y)$ and $\theta(xy) = \theta(x)\theta(y)$

Definition.

Let R and S be rings. An **isomorphism** from R to S is a mapping

$$\theta : R \to S$$

that is **injective**, **surjective**, and a **ring homomorphism**.

If such isomorphism exist we write $R \cong S$, and say that R and S are **isomorphic**.

Next Example

- We have shown that complex conjugation

$$f : \mathbb{C} \to \mathbb{C} \quad \text{given by } f(a + ib) = a - ib$$

  is a ring homomorphism.

- Next, we will show it is not just a ring **homomorphism**, but also a ring **isomorphism**.

- To do this, we need to prove f is both
  - injective, and
  - surjective.

Example 1

Example

Today: The complex conjugation map $f : \mathbb{C} \to \mathbb{C}$ given by $f(a+ib) = a - ib$ is a ring homomorphism.

Let us check whether f is a ring **isomorphism**.

- **Injective:** Suppose $f(a+ib) = f(c+id)$, we must show

  $$a + ib = c + id.$$

  In fact, $f(a+ib) = f(c+id)$ is equivalent to $a - ib = c - id$.

  By comparing real and imaginary parts, we deduce that $a = c$ and $b = d$.

  In particular, $a + ib = c + id$.

Example 1 - Part 2

Example (continuation)

- **Surjective:** Given $a + ib \in C$, we must find $x \in \mathbb{C}$ such that

$$f(x) = a + ib.$$

  Note that $x = a - ib$ is the required element.

Consequently, f is a **ring isomorphism**.

Remark

- Since $\mathbb{C}$ is an integral domain, the map f is also an **integral domain isomorphism**.
- Since $\mathbb{C}$ is a field, the map f is also a **field isomorphism**.

Next Example

- Next, we will see an example of a ring homomorphism that is **not** a ring isomorphism.
- More specifically, we will revisit the map $f : \mathbb{Z}/6\mathbb{Z} \to \mathbb{Z}/6\mathbb{Z}$ given by

$$f(x) = \overline{4}x, \quad \text{for all} \quad x \in \mathbb{Z}/6\mathbb{Z},$$

  which we showed last week to be a ring homomorphism.
- We will that it fails to be injective (and surjective).

Example 2

Example

Last week: The map $f : \mathbb{Z}/6\mathbb{Z} \to \mathbb{Z}/6\mathbb{Z}$ given by

$$f(x) = \overline{4}x, \quad \text{for all} \quad x \in \mathbb{Z}/6\mathbb{Z}$$

is a ring homomorphism.

Let us check whether f is a ring isomorphism.

Check injectivity: If $f(x) = f(y)$, we must check whether $x = y$.

Now, $f(x) = f(y)$ implies $\overline{4}x = \overline{4}y$.

Because $\mathbb{Z}/6\mathbb{Z}$ is **not an integral domain**, cancellation laws **do not hold!** For instance, we have $\overline{4} \cdot \overline{4} = \overline{16} = \overline{4}$ and $\overline{4} \cdot \overline{1} = \overline{4}$.

Thus, $f(\overline{4}) = f(\overline{1})$. We conclude f is **not** a ring isomorphism.

Remark

- The map from the previous example is **not** surjective.
- Here are two ways to see this:
  - For finite sets, a map $f : \mathbb{Z}/6\mathbb{Z} \to \mathbb{Z}/6\mathbb{Z}$ is surjective if and only if it is injective.

    Since f is not injective, it cannot be surjective.
  - Using the definition: For f to be surjective, every $y \in \mathbb{Z}/6\mathbb{Z}$ must have an $x \in \mathbb{Z}/6\mathbb{Z}$ such that $f(x) = y$.

    However,
    $$f(\overline{0}) = \overline{0}, \quad f(\overline{1}) = \overline{4}, \quad f(\overline{2}) = \overline{2},$$
    $$f(\overline{3}) = \overline{0}, \quad f(\overline{4}) = \overline{4}, \quad f(\overline{5}) = \overline{2}$$

    So, there is no $x \in \mathbb{Z}/6\mathbb{Z}$ such that $f(x) = \overline{1}$, $\overline{3}$, or $\overline{5}$.

Next slides: isomorphic vs non-isomorphic rings

- As already mentioned, if there is a ring isomorphism $\varphi : R \to S$, then R and S are essentially the same ring.
- In particular, R and S being isomorphic means that they share the same properties.
- So, if two rings differ with respect to a property, they are not isomorphic.
- In the next slides, we will use this to show that certain rings are not isomorphic.
- After that, we will explore properties that are shared by isomorphic rings.

Isomorphic vs non-isomorphic rings

- Rings R and S are isomorphic if there is a ring isomorphism $\varphi : R \to S$.
- They are **not** isomorphic if no such isomorphism exists.

Proving isomorphism vs non-isomorphism

- To show $R \cong S$, find a ring isomorphism $\varphi : R \to S$.
- To show $R \ncong S$, prove no such isomorphism exists.
- Checking all possible maps $\varphi : R \to S$ is too much work!
- Instead, use ring theoretical properties.

Non-isomorphic rings

Recipe for proving rings that are not isomorphic

- Find some property that one ring has, but that the other ring does not.
- Conclude they cannot be isomorphic.

Examples

- $\mathbb{Z}$ and $\mathbb{Z}_n$ are rings. However, $\mathbb{Z}$ is infinite and $\mathbb{Z}_n$ is finite. Thus, there is no isomorphism between them.
- $\mathbb{Z}$ and $\mathbb{Q}$ are not isomorphic because $\mathbb{Q}$ is a field, but $\mathbb{Z}$ is not.
- $\mathbb{Z}_3$ and $\mathbb{Z}_8$ are rings. However, $\mathbb{Z}_3$ is a field, while $\mathbb{Z}_8$ is not even an integral domain. Thus, they are not isomorphic.
- $\mathbb{Z}$ and $2\mathbb{Z}$ are both rings. However, $\mathbb{Z}$ has unity and $2\mathbb{Z}$ has not. Thus, they are not isomorphic.

Properties preserved by ring isomorphisms

If one of two **isomorphic rings**

- is **commutative**, then so is the other,
- **has unit**, then so does the other,
- **has a zero divisor**, then so does the other,
- is an **integral domain**, then so is the other,
- is a **field**, then so is the other.

Ideals

Next slides: ideals

- Roughly speaking, an ideal is a subring $I \subseteq R$ that 'absorbs' the elements of R under multiplication.
- Ideals are fundamental in ring theory because they generalise the concept of even numbers in the integers.
- They also provide a way to construct new rings, called quotient rings.

Definition.

A subring I of a ring R is an **ideal** of R if

$$ar \in I \text{ and } ra \in I$$

for all $a \in I$ and all $r \in R$.

We write $I \lhd R$ to indicate that I is an ideal of R.

Example

We know that $2\mathbb{Z} \subset \mathbb{Z}$ is a subring. Now, given $2z \in 2\mathbb{Z}$ and $x \in \mathbb{Z}$, we have

$$2z \cdot x = 2(zx) \in 2\mathbb{Z} \quad \text{and} \quad x \cdot 2z = 2(xz) \in 2\mathbb{Z}$$

because $\mathbb{Z}$ and $2\mathbb{Z}$ are commutative rings. Thus, $2\mathbb{Z} \lhd \mathbb{Z}$.

Non-example

Although $\mathbb{Z}$ is a subring of $\mathbb{Q}$, it is **not an ideal**: setting $a = 1$ and $r = 1/2$ yields

$$ar = 1/2 \notin \mathbb{Z}.$$

The trivial examples

For every ring R:

- $\{0\} \lhd R$ because $0r = r0 = 0$, for all $r \in R$,
- $R \lhd R$ because R is closed with respect to its multiplication.

Important example: polynomial ring

Recall: $\mathbb{R}[x]$

Recall that $\mathbb{R}[x]$ is the ring of all polynomials

$$a_0 + a_1 x + \cdots + a_n x^n, \text{ with } a_i \in \mathbb{R} \text{ and } n \in \mathbb{N} \cup \{0\}.$$

Example.

The following subset of $\mathbb{R}[x]$ is an ideal:

$$\mathcal{F} = \{(x^2+1)f(x) \mid f(x) \in \mathbb{R}[x]\}.$$

(Notice that $\mathcal{F}$ is the set of all polynomials in $\mathbb{R}[x]$ that are divisible by x^2+1.)

Let us show $\mathcal{F} \lhd \mathbb{R}[x]$

One can show $\mathcal{F}$ is a subring of $\mathbb{R}[x]$ (Exercise!).

Given $a = (x^2+1)f(x) \in \mathcal{F}$ and $r = g(x) \in \mathbb{R}[x]$, we must show

$$ar, ra \in \mathcal{F}.$$

We have

$$ar = \left((x^2+1)f(x)\right)g(x) = (x^2+1)(fg)(x) \in \mathcal{F}.$$

Similary, $ra \in \mathcal{F}$.

Next slides: image and Kernel

- We learned about the image and the kernel of a group homomorphism.
- Similarly, we can define the image and kernel of a **ring homomorphism**.
- This is crucial for proving an important theorem.
- The theorem states: The image of a ring homomorphism $\varphi : R \to S$ is a subring of S, and the kernel is an ideal of R.
- This provides a new way to find subrings and ideals of rings.

Image & Kernel

Given a **ring homomorphism** $\theta : R \to S$, the **image** of θ is

$$\mathrm{Im}(\theta) = \{\theta(r) \mid r \in R\} \subset S,$$

and the **kernel** of θ is

$$\mathrm{Ker}(\theta) = \{r \in R \mid \theta(r) = 0_s\} \subset R.$$

Theorem.

Let $\theta : R \to S$ be a ring homomorphism.

1. The image of θ is a subring of S.
2. The kernel of θ is an ideal of R.
3. θ is injective if and only if $\ker \theta = \{0_R\}$.

Proof.

Next Practical Session:

1. The image of θ is a subring of S.
2. The kernel of θ is an ideal of R.

Let us then show the third property.

3. θ is injective if and only if $\ker \theta = \{0_R\}$.

Proof of Theorem

θ is injective if and only if $\ker\theta = \{0_R\}$.

($\Longrightarrow$) Suppose θ is injective. Let us show $\ker(\theta) = \{0_R\}$.

Suppose $r \in \ker(\theta)$. This means that $\theta(r) = 0_S$.

We know from properties of homomorphisms that $\theta(0_R) = 0_S$.

Since θ is injective,

$$\theta(r) = 0_S = \theta(0_R) \;\text{ implies } r = 0_R.$$

This means that the only element of $\ker(\theta)$ is 0_R.

Proof of Theorem–Part 2

θ is injective if and only if $\ker\theta = \{0_R\}$.

($\Longleftarrow$) Suppose that $\ker(\theta) = \{0_R\}$. Let us show that θ is injective.

Assume $\theta(r) = \theta(s)$. We must show $r = s$. Notice that

$$\theta(r-s) = \theta(r+(-s)) = \theta(r) + \theta(-s) = \theta(r) - \theta(s) = 0_S.$$

By definition, this means that

$$r - s \in \ker(\theta).$$

Since $\ker(\theta) = \{0_R\}$, we see that $r - s = 0_R$. Hence $r = s$. $\square$

Remark

The previous Theorem can be used to find ideals!

Let us see an example.

Example

Last week: The map $f : \mathbb{Z}/6\mathbb{Z} \to \mathbb{Z}/6\mathbb{Z}$ given by

$$f(x) = \overline{4}x, \quad \text{for all} \quad x \in \mathbb{Z}/6\mathbb{Z}$$

is a ring homomorphism.

By definition,

$$\mathrm{Ker}(f) = \{z \in \mathbb{Z}/6\mathbb{Z} \mid f(z) = \overline{0}\}.$$

So, we need to find all solutions to $f(z) = \overline{0}$. Since $\mathbb{Z}/6\mathbb{Z}$ has only six elements, we can compute all $f(z)$.

Example

We have

$$f(\overline{0}) = \overline{4} \cdot \overline{0} = \boxed{\overline{0}}, \qquad f(\overline{1}) = \overline{4} \cdot \overline{1} = \overline{4}, \qquad f(\overline{2}) = \overline{4} \cdot \overline{2} = \overline{8} = \overline{2},$$
$$f(\overline{3}) = \overline{4} \cdot \overline{3} = \boxed{\overline{0}}, \qquad f(\overline{4}) = \overline{4} \cdot \overline{4} = \overline{4}, \qquad f(\overline{5}) = \overline{4} \cdot \overline{5} = \overline{2}$$

Thus,

$$\mathrm{Ker}(f) = \{\overline{0}, \overline{3}\}.$$

Next lecture

Next time...

- Ideals,
- Principal Ideals,
- Quotient Rings,
- Field of Fractions.